

10.9 (a) Following the result below Eq. (10.31), we have

$$\frac{\vec{E}^* \cdot \vec{A}_{sc}}{D_0} = k^2 \frac{\partial \mathcal{E}}{\partial \epsilon_0} (\vec{E}^* \cdot \vec{E}_0) \left[\frac{\sin qa - qa \cos qa}{q^3} \right] = k^2 a^3 (\epsilon_r - 1) (\vec{E}^* \cdot \vec{E}_0) \frac{j_1(qa)}{qa}$$

where $\vec{q} = k(\vec{n}_0 - \vec{n})$. The differential cross section is

$$\frac{d\sigma}{d\Omega} = \left| \frac{\vec{E}^* \cdot \vec{A}_{sc}}{D_0} \right|^2 = k^4 a^6 |\epsilon_r - 1|^2 |\vec{E}^* \cdot \vec{E}_0|^2 \left| \frac{j_1(qa)}{qa} \right|^2$$

Following Section 10.1 and Figure 10.1, we know $q = 2k \sin \theta/2$, and upon averaging over polarization, the total cross section becomes

$$\sigma = k^4 a^6 |\epsilon_r - 1|^2 \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \frac{1}{2} (1 + \cos^2 \theta) \left| \frac{j_1(qa)}{qa} \right|^2$$

For $ka \gg 1$, the major contribution of the above integral comes from the region $\theta \approx 0$. Linearize q , we can approximate the integral as

$$\sigma = 2\pi k^4 a^6 |\epsilon_r - 1|^2 \int_0^\pi d\theta \theta \left(\frac{j_1(ka\theta)}{ka\theta} \right)^2$$

$$\rightarrow 2\pi k^4 a^6 |\epsilon_r - 1|^2 \frac{1}{k^2 a^2} \int_0^{+\infty} \frac{j_1(x)^2}{x} dx = \frac{\pi}{2} k^2 a^4 |\epsilon_r - 1|^2$$

where we have used the result that

$$\int_0^{+\infty} \frac{j_1(x)^2}{x} dx = \frac{1}{4}$$

(b) For arbitrary ka , we need to perform the integral in part (a) without approximation.

$$\sigma = \frac{\pi}{2} k^2 a^4 |\epsilon_r - 1|^2 \cdot 2k^2 a^2 \int_0^\pi d\theta \sin \theta (1 + \cos^2 \theta) \frac{(\sin qa - qa \cos qa)^2}{(qa)^6}$$

Let $x = 2ka \sin \theta/2$, then $dx = ka \cos \frac{\theta}{2} d\theta$,

$$\sin \theta d\theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} d\theta = \frac{x}{ka} \frac{dx}{ka} = \frac{1}{(ka)^2} x dx$$

$$1 + \cos^2 \theta = 1 + (1 - 2\sin^2 \frac{\theta}{2})^2 = 2 - 4(\sin \frac{\theta}{2})^2 + 4(\sin \frac{\theta}{2})^4 = 2 - \frac{x^2}{k^2 a^2} + \frac{x^4}{4k^2 a^2}$$

and the factor becomes

$$F = 2 \int_0^{2ka} \frac{(\sin x - x \cos x)^2}{x^6} \left(2 - \frac{x^2}{k^2 a^2} + \frac{x^4}{4k^2 a^2} \right) x dx$$